Homework — 2.1.5 Thinking Concurrently

Name: ______________________ Date: __________ Mark: ______ / 20

OCR H446 — set after the 2.1.5 lesson.

Part A — This week's topic (~14 marks)

1. When a customer books a trip on a holiday-booking website, the system must: charge the customer's card, reserve the flight seat, reserve the hotel room, and send a confirmation email. Complete the table: for each pair of tasks, state whether they could run at the **same time** or must be **sequenced**, and give the reason. **[3]** *(AO2)*

Task pair	Same time or sequenced?	Reason
Reserve the flight seat / reserve the hotel room		
Charge the customer's card / send the confirmation email		
Reserve the flight seat / send the confirmation email		

2. Write the letter of the correct definition in the Match column for each term. Two definitions are not needed. **[2]** *(AO1)*

Term	Match
Concurrent processing	
Race condition	
Data dependency	
Multi-core processor	

Letter	Definition
A	More than one task making progress during the same period of time (on a single core this is achieved by time-slicing)
B	Hardware containing several processing units, allowing tasks to execute genuinely in parallel
C	A flaw where the final result depends on the unpredictable order in which tasks access shared data
D	Where one task needs the output of another before it can start, forcing the tasks to be sequenced
E	Breaking a complex problem down into smaller sub-problems
F	A task deliberately left to run only when the system is otherwise idle

3. For each statement, write **True** or **False** in the middle column. If a statement is false, write a corrected version in the final column. **[4]** *(AO2)*

Statement	True / False	Correction if false
Running a program's tasks concurrently will always make the program finish faster.		
On a single-core processor, concurrent tasks are time-sliced rather than executed truly in parallel.		
Rendering the 200 individual frames of an animated film is poorly suited to concurrent processing, because each frame depends on the result of the previous frame.		
Creating, coordinating and recombining concurrent tasks adds an overhead that can outweigh the speed gain.		

4. Two members of staff use the same warehouse stock system at the same instant. Both read "stock = 1" for the last remaining item and both confirm a sale, so the item is sold twice. Complete the paragraph using words from the word bank. Three words are not needed. **[2]** *(AO2)*

Both processes read the same **(a)** __________ stock value before either had written its update. Because the two updates were not coordinated, the final result depended on the unpredictable **(b)** __________ of the operations, so the last item was sold **(c)** __________ . This problem is called a **(d)** __________ condition.

Word bank: shared · timing · twice · race · local · once · dependency

5. When the user of a word processor clicks "Save", the program also runs a long spell-check in the background so the user can keep typing.

(a) Explain **one** benefit of running the spell-check **concurrently** with editing. **[2]** *(AO2)*

(b) State **one** hardware feature that would allow the two tasks to be processed genuinely in parallel. **[1]** *(AO1)*

Part B — Retrieval: keep it fresh (~6 marks)

6. *(2.1.4)* A game ends a player's turn when they have **either** run out of moves **or** scored 100 points or more. Write a suitable Boolean **condition** for this decision point using a logical operator. **[2]** *(AO2)*

7. *(1.4.3)* Complete the truth table for `Q = A AND (NOT B)`. **[2]** *(AO2)*

A	**B**	**Q**
0	0	
0	1	
1	0	
1	1	

8. *(1.4.2)* State **one** difference between a **static** and a **dynamic** data structure. **[2]** *(AO1)*